

Orlane Rossini

✉ orlane.rossini@umontpellier.fr     orlross.github.io

Education

PhD in Applied Mathematics with specialisation in Biostatistics 2022 - November 2025
Institut Montpelliérain Alexander Grothendieck (IMAG) Montpellier, France

- **Thesis subject:** Model-based reinforcement learning for the control of partially observable piecewise deterministic semi-Markov processes
- **Supervisors:** Benoîte de Saporta, Alice Cleynen and Régis Sabbadin

Master in Data Science & Statistical Modeling 2020 - 2022
University of South Brittany Vannes, France

Academic experience

Temporary Lecturer Sep 2025
Department of Mathematics, Faculty of Science, University of Montpellier Montpellier, France

- Lectures, tutorials and lab session for Bachelor and Master students

Assistant Lecturer Oct 2022 – June 2025
Department of Mathematics, Faculty of Science, University of Montpellier Montpellier, France

- Tutorials and lab session for Bachelor and Master students

Student Intern, then Research engineer Jan 2022 – Sep 2022
University of Grenoble, INSERM, IAB Grenoble, France

- **Subject:** Accelerated epigenetic placental aging as a biomarker of pregnancy exposure to ambient air pollution
- **Supervisors:** Johanna Lepeule, Aurélie Nakaruma, Lucile Broséus and François Septier

Mentored project Jan 2021 – May 2021
Regional Health Observatory of French Guiana Cayenne, France

- **Subject:** Analysis of patient flow in the emergency department and its associated factors in French Guiana between 2018 and 2020
- **Supervisors:** Joseph Rwagitinywa and Evans Gouno

Publications

Preprint

- *Bayes-Adaptive Impulse Control of Piecewise-Deterministic Markov Processes* in hal-05054392 by O. Rossini, M. Vinyals, A. Cleynen, B. de Saporta, R. Sabbadin, M. Vinyals October 2025
- *Bridging Impulse Control of Piecewise Deterministic Markov Processes and Markov Decision Processes: Frameworks, Extensions, and Open Challenges* in arXiv:2501.04120 by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin, A. Vernay April 2025

In Proceedings of Conferences

- *Deep Reinforcement Learning for Controlled Piecewise Deterministic Markov Process in Cancer Treatment Follow-up* in Journée Santé et IA by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin, M. Vinyals July 2024
- *Deep reinforcement learning for controlled piecewise deterministic Markov process in cancer treatment follow-up* in Journées des Statistiques de France by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin, M. Vinyals May 2024
- *An example of medical treatment optimization under model uncertainty* in Journées des Statistiques de France by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin. July 2023

Supervision

Co-supervision (50% with A. Vernay, PHD fellow IMAG, Montpellier) of a 1st year observation internship Khalil Elhajoui

- **Subject:** Introduction to Supervised Learning and the Use of Perceptrons

Scientific mediation

- Workshop at the *Fête de la Science* "Mathematics with head and hands" (2022)
- Workshop at the *Girls, Math and Computer Science* event (2023)
- Workshop at the *Math en Jeans* conference (2024)

Administratives

- Elected member of the IMAG laboratory council (from 1st February 2024)
- Member of the sustainable development group at the IMAG laboratory (from January 2024 to January 2025)
- Member of the organising committee of the *Math en Jeans* conference (2024)

Talks

International conferences

- *Deep Reinforcement Learning for Impulse Control in PDMPs through BAPOMDP framework*, MaSeMo : Markov, Semi-Markov Models and Associated Fields, July 2025
- *An example of medical treatment optimization under model uncertainty*, 21st INFORMS Applied Probability Society Conference, June 2023
- *Maternal exposure to ambient air pollution and placental epigenetic aging: Developing a new clock and estimating associations*, Australian Mathematical Society, December 2022

National conferences

- *Deep Reinforcement Learning for Bayes-Adaptive Impulse Control of PDMPs for Cancer Patient Follow-Up*, Séminaire de Probabilité et Statistiques de l'IMAG, Montpellier, October 2025
- *Predicting Yeast Translation Efficiency from RNA Modification and Degradation Features: A Machine Learning Approach*, Journées de Biostatistiques, November 2024
- *Deep reinforcement learning for piecewise deterministic Markov process control*, Rencontres Montpellier-Sherbrooke, June 2024
- *Stochastic dynamic control: an approach based on semi-Markov models*, 10èmes Rencontres des Jeunes Statisticien.ne.s, April 2024
- *Stochastic dynamic control: an approach based on semi-Markov models, application to the optimization of a medical treatment*, 8èmes rencontres de Statistique of University of South Brittany, November 2023
- *An example of medical treatment optimization under model uncertainty*, Processus markoviens, semi-markoviens et leurs applications, June 2023
- *Maternal exposure to ambient air pollution and placental epigenetic aging: Developing a new clock and estimating associations*, IMAG PhD student days, February 2023
- *Model-based reinforcement learning for the control of partially observable, piecewise deterministic semi-Markovian decision processes*, INRAE-MIAT PhD student days, January 2023